

Week 15 (8.29-9.3)

Module 1:

(A) Verification of Expression Efficiency

1. Inoculation

Original strains: BW-pYB1a-*progo*, BW-pYB1a-*nov1f*, BW-pYB1a-*progo-nov1f*, BW-pYB1a-*vdh*, BW-pYB1a-*vdh-nox*

Cultivation in a shaker at 37°C for 12 hours

2. Induction:

Induction of BW-pYB1a-*progo*, BW-pYB1a-*nov1f*, BW-pYB1a-*progo-nov1f*, BW-pRB1k-*vdh*, BW-pRB1k-*vdh-nox*

Induction System Component	Volume
ZY	4.8 mL
50x 5052 supplement	100 μ L
50x M supplement	100 μ L
MgSO ₄	10 μ L
1000x trace element	10 μ L
Bacterial suspension	50 μ L
arabinose	50 μ L
Antibiotic	5 μ L

cultivation in a shaker at 30°C for 16 hours.

3. Whole-Cell Catalysis:

Using 4-PG to catalyze the induced enzymes for Isoeugenol production and Isoeugenol to catalyze the induced enzymes for vanillin production, vanillin to catalyze the induced enzymes for VA production

Determine the OD₆₀₀ of induced bacterial cells using an ultraviolet spectrophotometer.

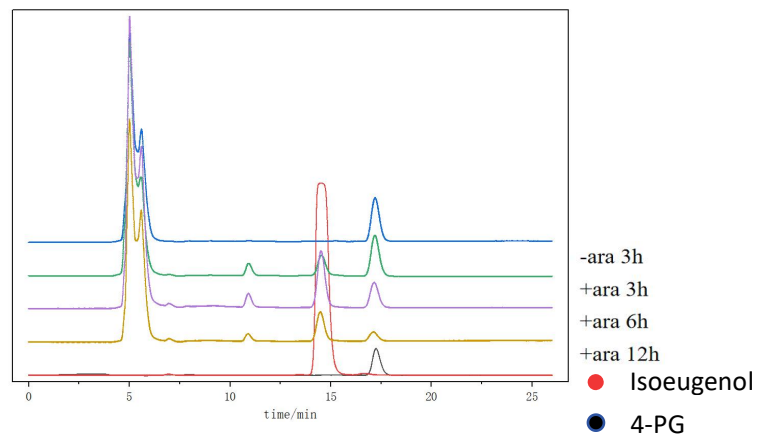
Collect 6 OD units of bacterial cells into a centrifuge tube, centrifuge at 4200 rpm for 10 minutes.

Discard the supernatant, resuspend the bacterial cells in 50mM Tris-HCl(pH=7.5), and transfer to a centrifuge tube for whole-cell catalysis.

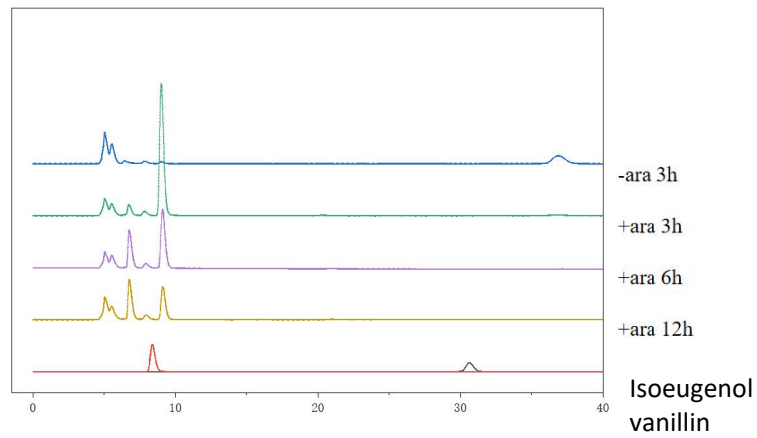
Incubate in a shaker at 30°C for 1h, 3h, 6h, 9h, 12h hours.

4. High-Performance Liquid Chromatography (HPLC)

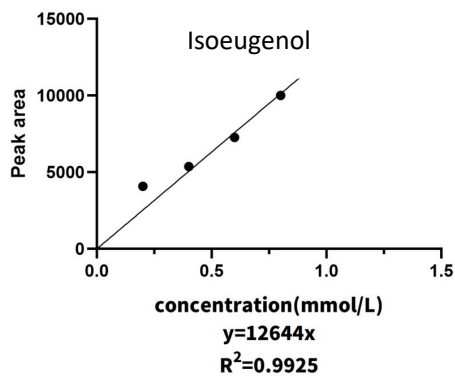
Detection of Isoeugenol



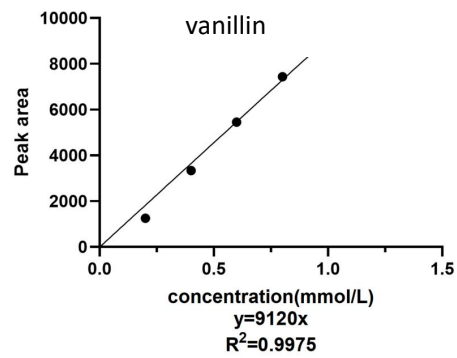
Detection of vanillin



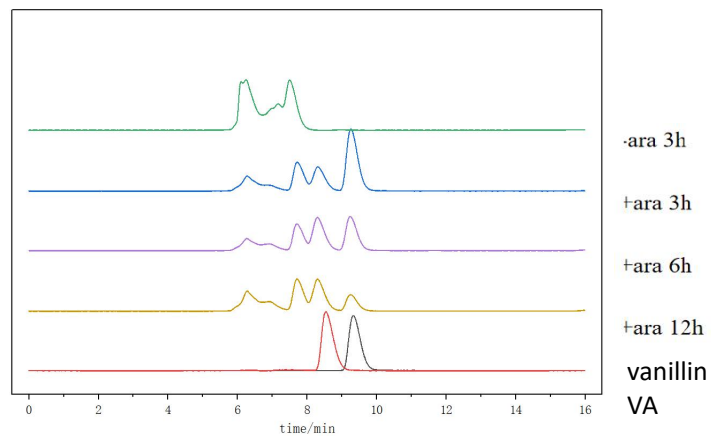
异丁香酚标准曲线

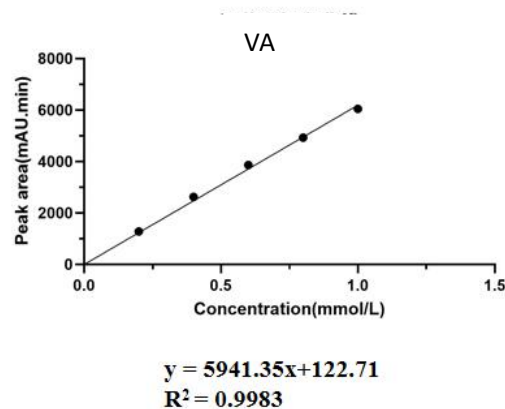
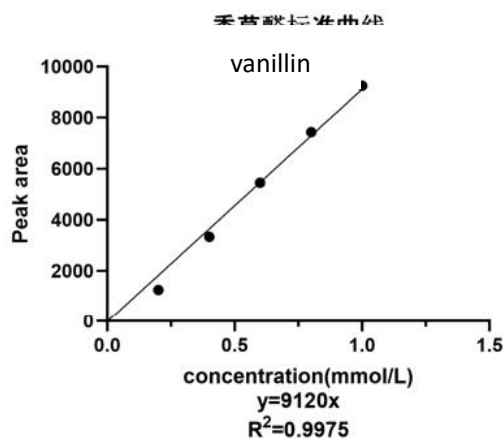


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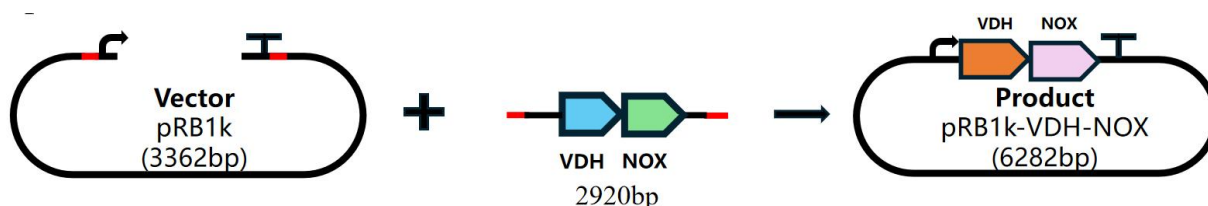


Detection of VA





(B) Construct BW-pRB1k-*vdh-nox*



PCR + cut gel recovery

The primer was designed, followed by PCR amplification of the target gene and enzymatic digestion of vector pRB1k

PCR system (50μL)	
2×Mix	25ul
<i>vdh-nox</i> -F	2ul
<i>vdh-nox</i> -R	2ul
Template	20ng
DDW	To 50ul

PCR system (50μL)	
2×Mix	25ul
pRB1k-F	2ul
pRB1k-R	2ul
Template	20ng
DDW	To 50ul

PCR (<i>vdh-nox</i>)	
98°C	5min
98°C	30s
64°C	30s
72°C	2min45s
72°C	5min
25°C	∞

PCR (pRB1k)	
98°C	5min
98°C	30s
64°C	30s
72°C	1min45s
72°C	5min
25°C	∞

The viscosity of the adhesive recovery is too low.

Module 2:

1. Isolation and inoculation of single colonies

Single colonies were isolated by streaking on agar from M9 culture solutions after three rounds of transfer and screening, which were then inoculated into LB medium supplemented with 1% antibiotic for cultivation to obtain bacterial liquid.

2. Inoculation of control groups

Original strains: BW Δ *codA*-pYB1a-*odm*^{WT}

Cultivation in a shaker at 37°C for 12 hours

3. Induction of BW Δ *codA*-pYB1a-*odm*^{WT}, and selected isolated colonies.

Induction System Component	Volume
ZY	4.8 mL
50x 5052 supplement	100 μ L
50x M supplement	100 μ L
MgSO ₄	10 μ L
1000x trace element	10 μ L
Bacterial suspension	50 μ L
20% arabinose	50 μ L
Antibiotic	5 μ L

Cultivation in a shaker at 30°C for 16 hours.

4. Whole-Cell Catalysis:

Using 20% arabinose to catalyze the induced enzymes for 4-HB production. Determine the OD₆₀₀ of induced bacterial cells using an ultraviolet spectrophotometer. Collect OD₆₀₀=8 units of bacterial cells into a centrifuge tube, centrifuge at 4200 rpm for 10 minutes. Discard the supernatant, resuspend the bacterial cells in 2.5 mM VA at a final concentration, with Tris-HCl (pH = 7.5) as the solvent, and transfer to a centrifuge tube for whole-cell catalysis. Incubate in a shaker at 30°C for 12 hours. After catalysis, the system was centrifuged, filtered through 0.22 μ m membranes, and stored at -20°C.

5. High-Performance Liquid Chromatography (HPLC) Detection of 4-HB

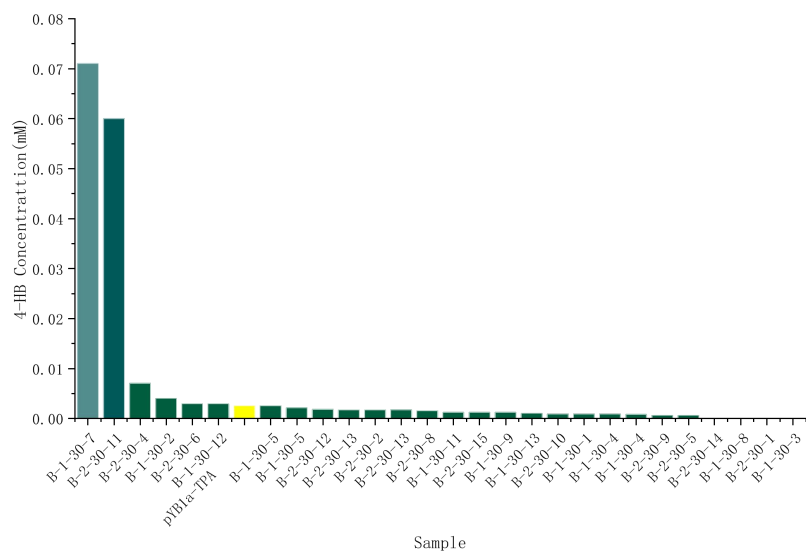
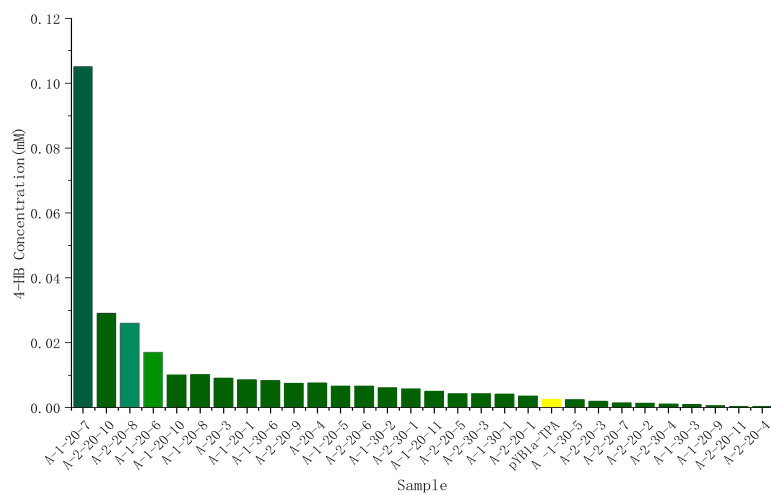
Prepare mobile phase, standards and filter samples

The mobile phase was prepared: 0.1% aqueous phosphoric acid : methanol = 75:25 (v/v)

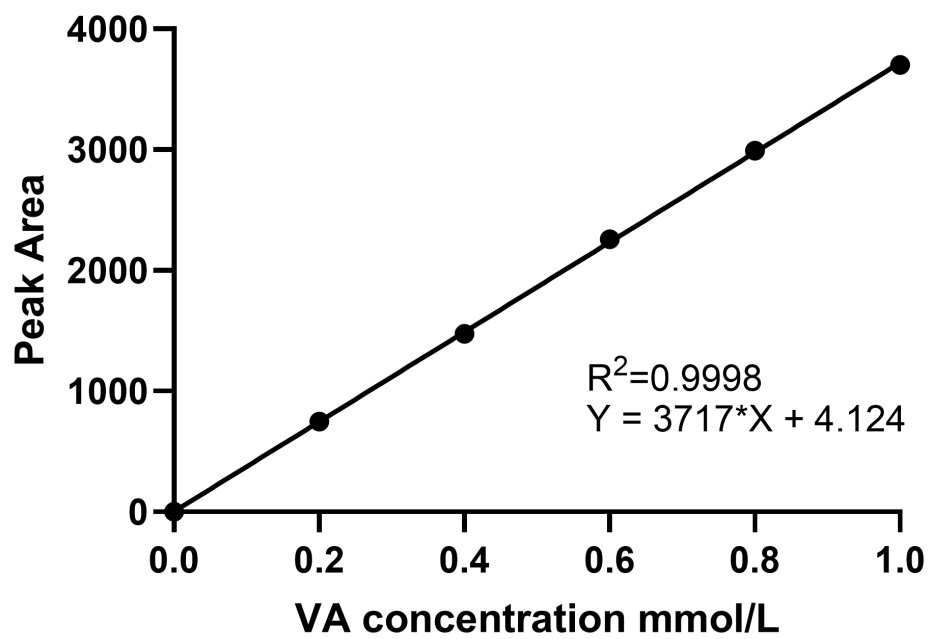
The standard solutions were prepared, and the samples were filtered through 0.22 μ m membranes for HPLC.

The samples were detected by HPLC using the mobile phase of 0.1% aqueous phosphoric acid : methanol = 75:25 (v/v) . Column: C18, 250nm $\times$ 4.6mm $\times$ 5 μ m; flow rate: 0.7 mL/min; column injection volume: 5 μ L; detection wavelength: 254nm.

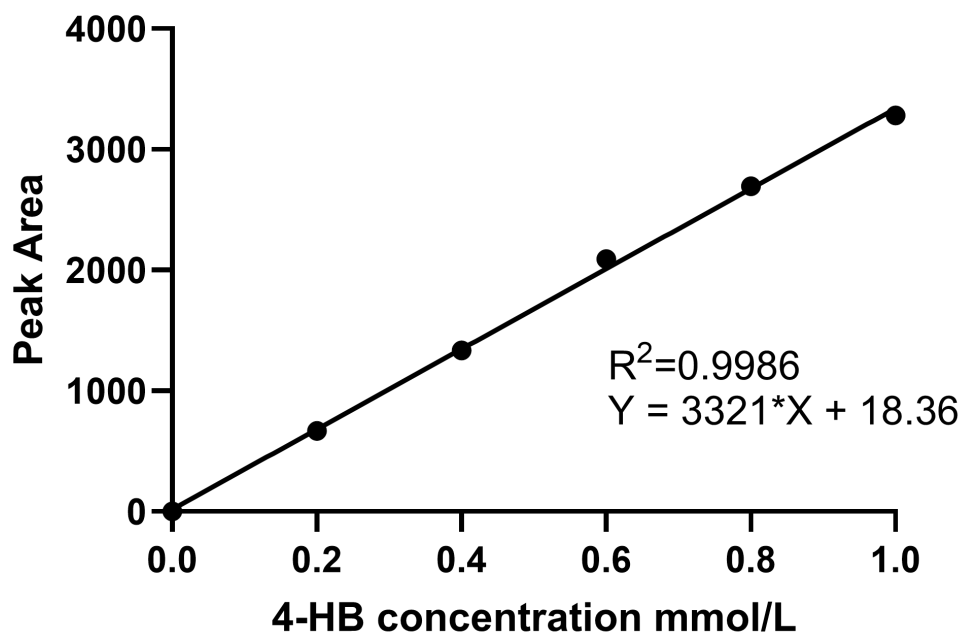
The samples were detected by HPLC using column: C18, 250nm $\times$ 4.6mm $\times$ 5 μ m; flow rate: 0.7 mL/min; column injection volume: 5 μ L; detection wavelength: 254nm.



VA



4-HB



High-performance strains exhibited the following yields:

A-1-20-7: 0.105 mM

A-2-20-10: 0.029 mM

A-2-20-8: 0.026 mM

B-1-30-7: 0.071 mM

B-2-30-11: 0.060 mM

These five high-performance strains showed **42.0-fold**, **11.6-fold**, **10.4-fold**, **28.4-fold** and **24.0-fold** product yields compared with the wild-type (pYB1a-*odm*^{WT}, 0.0025 mM). All concentrations above were calculated based on the standard curve.

Module 3:

(A)Run HPLC to verify pyruvate production.

1.Shake culture of strains No.1, 2 and 12, and send for testing

Strains No.1, No.2 and No.12 were inoculated and cultivated in LB medium (37°C, 12 hours), and then submitted for sequencing/testing.

2.Culture preservation and plasmid extraction

The strains were preserved (glycerol stock), and plasmids were extracted and their concentrations were measured.

3.Induction: Induction of BW-pLB1s-*VanAB-aroY*

Induction System Component	Volume
ZY	4.8 mL
50x 5052 supplement	100 µL
50x M supplement	100 µL
MgSO ₄	10 µL

1000x trace element	10 μ L
Bacterial suspension	50 μ L
arabinose	50 μ L
Antibiotic	5 μ L

cultivation in a shaker at 30°C for 16 hours.

4. Whole-cell catalysis

Measurement of the OD600 value.

BW-pYB1a-dmpB-dmpD-mhpD-mhpE using VA to catalyze the induced enzymes for Pyr production. BW-pLB1s-VanAB-aroY, BW-pYB1a-dmpB-dmpD-mhpD-mhpE using CA to catalyze the induced enzymes for Pyr production. Determine the OD 600 of induced bacterial cells using an ultraviolet spectrophotometer. Collect OD 600=6 units of bacterial cells into a centrifuge tube, centrifuge at 4200 rpm for 10 minutes. Discard the supernatant, resuspend the bacterial cells in H₂SO₄, and transfer to a centrifuge tube for whole-cell catalysis. Incubate in a shaker at 30 ° C for 1 hours.

The whole-cell catalytic system was configured as follows:

System	Substrate (final conc.)	Cofactors (final conc.)	Solvent
Uninduced Full pathway (BW-pLB1s-VanAB-aroY/BW-pYB1a-dmpBD-mhpDE)	VA 5mM (final)	/	5mM H ₂ SO ₄ solution
Uninduced Full pathway (BW-pLB1s-VanAB-aroY/BW-pYB1a-dmpBD-mhpDE)	VA 5mM (final)	NADH 10mM (final)	5mM H ₂ SO ₄ solution
Induced Full pathway (BW-pLB1s-VanAB-aroY/BW-pYB1a-dmpBD-mhpDE))	VA 5mM (final)	/	5mM H ₂ SO ₄ solution
Induced Full pathway (BW-pLB1s-VanAB-aroY/BW-pYB1a-dmpBD-mhpDE))	VA 5mM (final)	NADH 10mM (final)	5mM H ₂ SO ₄ solution
Uninduced BW	VA 5mM (final)	NADH 10mM (final)	5mM H ₂ SO ₄ solution
Induced BW	VA 5mM (final)	NADH 10mM (final)	5mM H ₂ SO ₄ solution
Uninduced BW-pYB1a-eGFP	VA 5mM (final)	NADH 10mM (final)	5mM H ₂ SO ₄ solution
Induced BW-pYB1a-eGFP	VA 5mM (final)	NADH 10mM (final)	5mM H ₂ SO ₄ solution
Uninduced Downstream pathway (BW-pYB1a-dmpBD-mhpDE)	CA 5mM (final)	/	5mM H ₂ SO ₄ solution
Uninduced Downstream pathway (BW-pYB1a-dmpBD-mhpDE)	CA 5mM (final)	Fe ²⁺ 0.5mM (final), Mn ²⁺ 1mM (final)	5mM H ₂ SO ₄ solution
Induced Downstream pathway (BW-pYB1a-dmpBD-mhpDE)	CA 5mM (final)	/	5mM H ₂ SO ₄ solution
Induced Downstream pathway (BW-pYB1a-dmpBD-mhpDE)	CA 5mM (final)	Fe ²⁺ 0.5mM (final), Mn ²⁺ 1mM (final)	5mM H ₂ SO ₄ solution
Uninduced BW	CA 5mM (final)	Fe ²⁺ 0.5mM (final), Mn ²⁺ 1mM (final)	5mM H ₂ SO ₄ solution

System	Substrate (final conc.)	Cofactors (final conc.)	Solvent
Induced BW	CA 5mM (final)	Fe2+ 0.5mM (final), Mn2+ 1mM (final)	5mM H2SO4 solution
Uninduced BW-pYB1a-eGFP	CA 5mM (final)	Fe2+ 0.5mM (final), Mn2+ 1mM (final)	5mM H2SO4 solution
Induced BW-pYB1a-eGFP	CA 5mM (final)	Fe2+ 0.5mM (final), Mn2+ 1mM (final)	5mM H2SO4 solution

All components were added to the cell pellet tubes and mixed by pipetting, then shaken at 30°C, 200r/min in the dark.

5.Prepare standards and mobile phase

The HPLC standard solutions and the mobile phase (5mM H2SO4 aqueous solution, etc.) were prepared.

6.HPLC (acetonitrile-ultrapure water method; 5mM dilute H2SO4 sugar-H column method)

The samples were detected by HPLC with the acetonitrile-ultrapure water method (mobile phases: 5% acetonitrile/5mM H2SO4, 30% acetonitrile/5mM H2SO4)

7.HPLC run.

The standards and the samples (full pathway and downstream pathway) were detected by HPLC with RID.

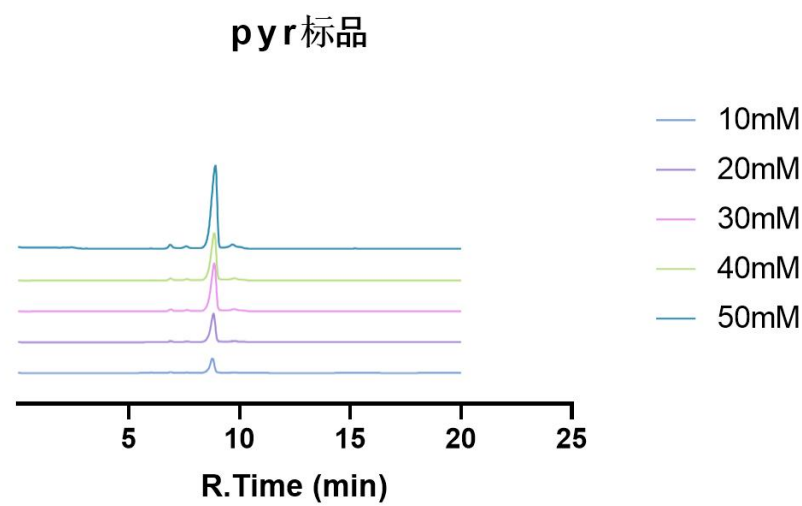


Figure 1: HPLC chromatograms of the pyruvate standards (10/8/6/4/2 mM).

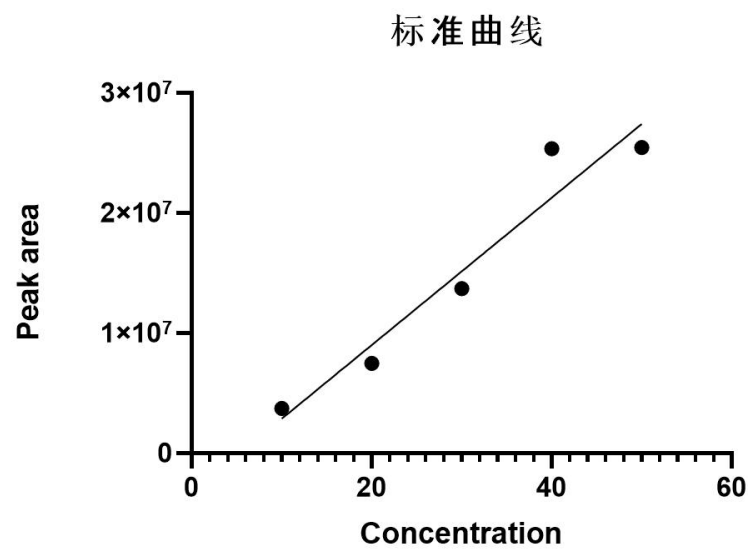


Figure 2: Standard curve of the pyruvate standards ($Y = 613501X - 3253881$, $R^2 = 0.9366$).

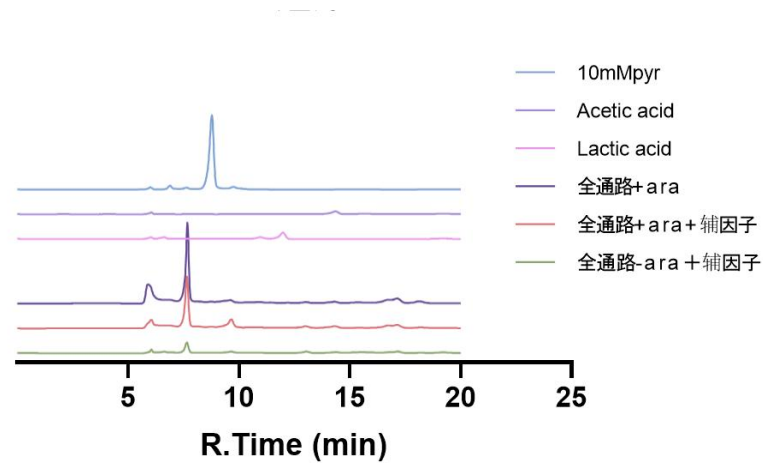


Figure 3: HPLC chromatograms of the full-pathway verification (pLB1s-VanAB-aroY/pYB1a-dmpBD-mhpDE/BW).

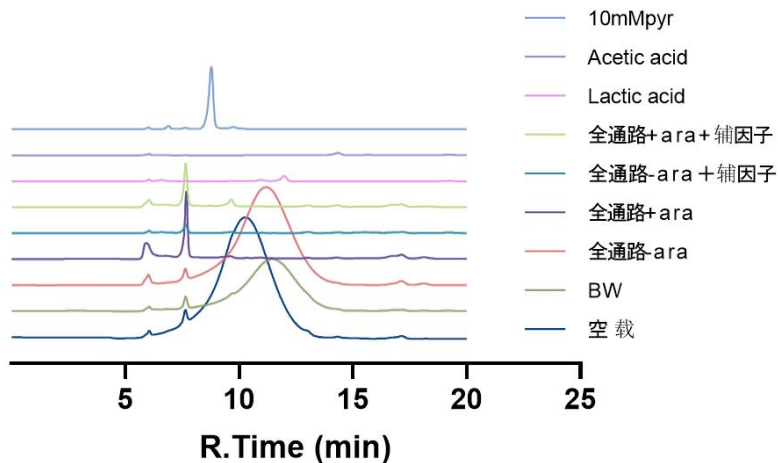


Figure 4: Overlay of the full-pathway samples with the pyruvate, acetic acid and lactic acid standards.

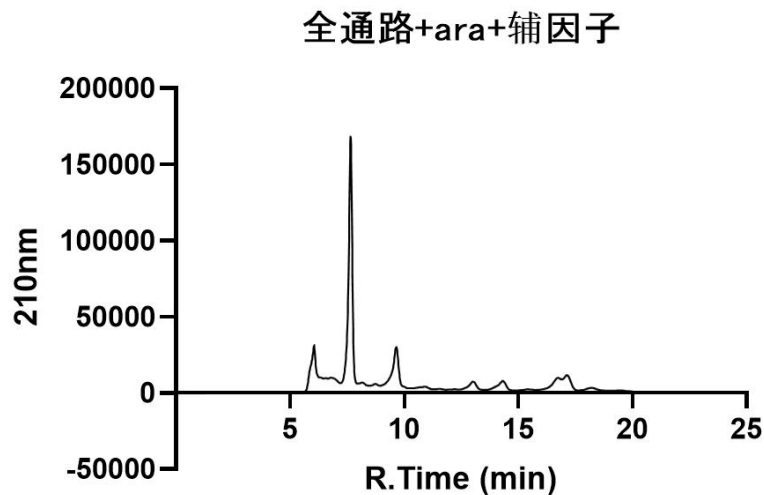


Figure 5: HPLC chromatogram of the full pathway +ara +cofactors (210 nm).

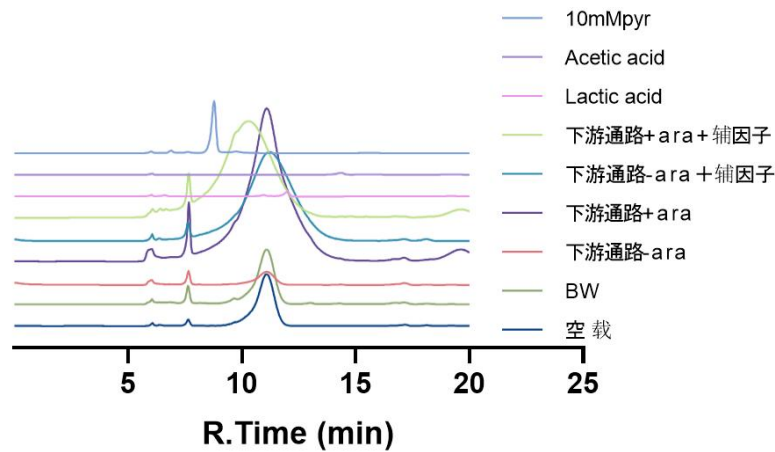


Figure 6: HPLC chromatograms of the downstream-pathway verification (pYB1a-dmpBD-mhpDE/BW).

(B)RBS library construction.

1.Re-PCR and gel recovery.

Re-PCR and gel recovery were carried out. Problem: the dmpD fragment could not be amplified; changing the vector, the annealing temperature and the extension time had all been tried.

2.PCR with the pYB1a-dmpD single-gene plasmid.

The pYB1a-dmpD single-gene plasmid was now used as the template for PCR, and the product was checked by running a recovery gel.

3.Sequencing of dmpD.

The pYB1a-dmpBD-mhpDE plasmid was submitted for sequencing (to detect dmpD); it may be a problem of the sequencing primers or an incorrect detection position. Re-PCR, gel running and sequencing of dmpD were repeated.

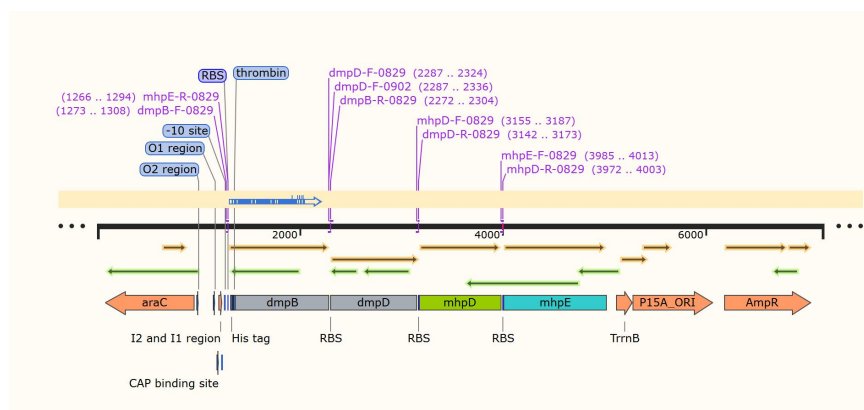


Figure 7: Primer design map for the RBS library (pYB1a-dmpB-dmpD-mhpD-mhpE(2), 7155 bp).

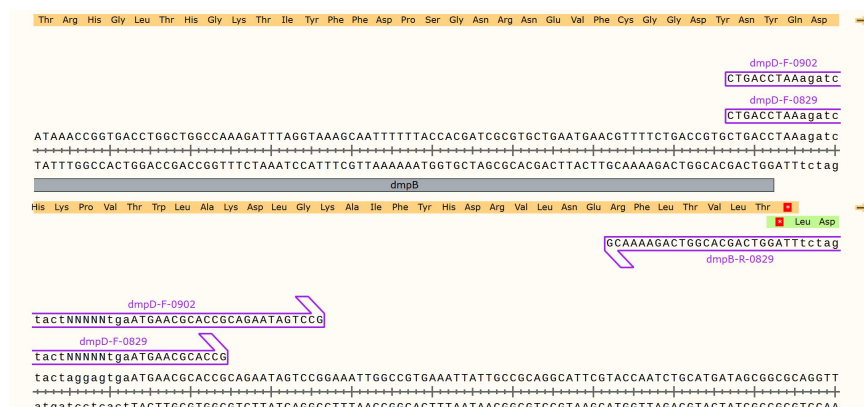


Figure 8: Sequence comparison around dmpD with the designed primers (dmpD-F-0902 / dmpD-F-0829 / dmpB-R-0829).

(C)Next week's plan.

1.Plans.

1. Continue the four-gene RBS library construction of pYB1a-dmpBD-mhpDE/BW; 2. Carry out the full-pathway HPLC verification of pLB1s-VanA-VanB-aroY/pYB1a-dmpBD-mhpDE/BW.

Week 16 (9.4-9.10)

Module 1:

(A)Construct pRB1k-*vdh-nox*

1.PCR + cut gel recovery

The primer was designed, followed by PCR amplification of the target gene and enzymatic digestion of vector pRB1k

Subsequently, the relevant DNA fragments were assembled using Gibson Assembly methodology, followed by transformation into competent cells. Upon completion of the transformation procedure, polymerase chain reaction (PCR) screening of bacterial colonies was performed.

PCR system (50μL)	
2×Mix	25ul
<i>vdh-nox</i> -F	2ul
<i>vdh-nox</i> -R	2ul
Template	20ng
DDW	To 50ul

PCR system (50μL)	
2×Mix	25ul
pRB1k-F	2ul
pRB1k-R	2ul
Template	20ng
DDW	To 50ul

PCR (<i>vdh-nox</i>)	
98°C	5min
98°C	30s
64°C	30s
72°C	2min45s
72°C	5min
25°C	∞

} × 30

PCR (pRB1k)	
98°C	5min
98°C	30s
64°C	30s
72°C	1min45s
72°C	5min
25°C	∞

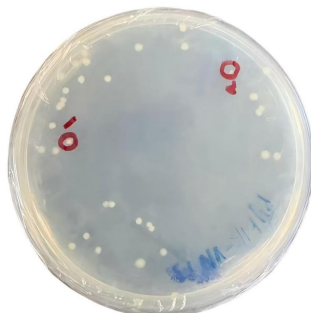
} × 30

2. Connection

Connecting the two fragments using the Gibson assembly method.

3.Transformation

The ligated system was transferred into DH5α receptor cells, coated with plates, and cultured inverted at 37°C for 12h;



4.Plasmid extraction + send for sequencing

The correctly digested protoplasmid was sent for testing, and pRB1k-*vdh-nox* sequenced correctly

(B)Construct a double recombinant plasmid containing pRB1k-*vdh-nox*
and pYB1a-*progo-nov1f*

Transformation

The plasmid pRB1k-*vdh-nox*and plasmid pYB1a-*progo-nov1f* was transferred into BW receptor cells,
coated with plates, and cultured inverted at 37°C for 12h;



(C) Induction and protein detection

1. Inoculation

Original strains: BW-pYB1a-*progo*、BW-pYB1a-*nov1f*、BW-pYB1a-*progo-nov1f*、
BW-pYB1a-*vdh*、BW-pYB1a-*vdh-nox*、BW-pRB1k-*vdh-nox*

Cultivation in a shaker at 37°C for 12 hours

2.Induction:

Induction of BW-pYB1a-*progo*、BW-pYB1a-*nov1f*、BW-pYB1a-*progo-nov1f*、
BW-pRB1k-*vdh*、BW-pRB1k-*vdh-nox*

Induction System Component	Volume
ZY	4.8 mL
50x 5052 supplement	100 μL
50x M supplement	100 μL
MgSO4	10 μL

1000x trace element	10 μ L
Bacterial suspension	50 μ L
arabinose	50 μ L
Antibiotic	5 μ L

cultivation in a shaker at 30°C for 16 hours.

3.1.1 Whole-Cell Catalysis:

Using vanillin to catalyze the induced enzymes for VA production

Determine the OD₆₀₀ of induced bacterial cells using an ultraviolet spectrophotometer.

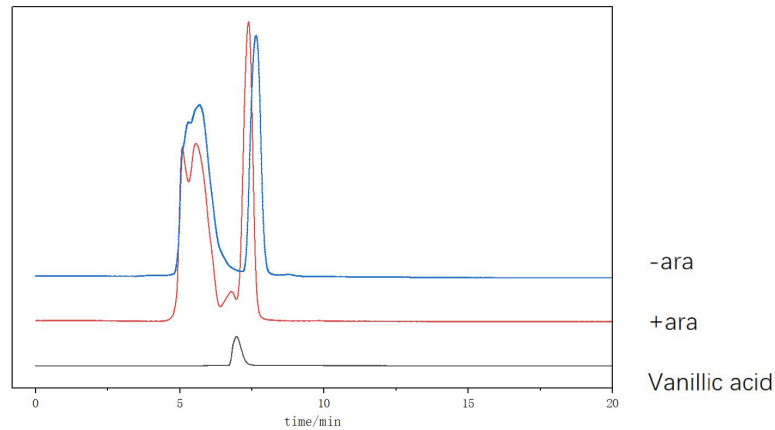
Collect 6 OD units of bacterial cells into a centrifuge tube, centrifuge at 4200 rpm for 10 minutes.

Discard the supernatant, resuspend the bacterial cells in 50mM Tris-HCl(pH=7.5), and transfer to a centrifuge tube for whole-cell catalysis.

Incubate in a shaker at 30°C for 6h hours.

3.1.2 High-Performance Liquid Chromatography (HPLC)

Detection of VA



3.2 Protein expression validation

1. Sample Preparation

The cultures were collected, and the bacterial suspension with a 6 OD unit was measured to obtain the final sample. After centrifugation at 10,000 g for 10 minutes, the supernatant was discarded, and the precipitate was resuspended in 600 μ L sterile water. The cells were lysed using an ultrasonic cell disruptor. The lysate was then centrifuged at 10,000 g for 10 minutes to separate the supernatant and sediment. Finally, the precipitate was resuspended in 600 μ L sterile water. This procedure completed the sample preparation.

2. point sample

Prepare the protein buffer solution according to the table below. Place the gel in the electrophoresis instrument. After checking for leaks with the buffer solution, mix 16 μ L of prepared sample with 4 μ L loading buffer. Transfer the mixed sample into the PCR instrument and heat at 100°C ° C for 5 minutes to denature proteins. After cooling, load 16 μ L

of the mixture into the gel well and add 4 μL of protein marker. The sample loading is now complete.

Concentrating Gel:

12% Lower Gel	2 Gel Units	3 Gel Units	4 Gel Units	5 Gel Units	6 Gel Units	8 Gel Units	10 Gel Units
Solution Components (mL)	10	15	20	25	30	40	50
Water	3.3	4.9	6.6	8.2	9.9	13.2	16.5
30% Acrylamide Solution	4	6	8	10	12	16	20
Tris (pH 8.8)	2.5	3.8	5	6.3	7.5	10	12.5
10% SDS	0.1	0.15	0.2	0.25	0.3	0.4	0.5
10% Ammonium Persulfate	0.1	0.15	0.2	0.25	0.3	0.4	0.5
TEMED	0.004	0.006	0.008	0.01	0.012	0.016	0.02

separation gel:

5% Stacking Gel	2 Gel Units	3 Gel Units	4 Gel Units	5 Gel Units	6 Gel Units	8 Gel Units	10 Gel Units	12 Gel Units
Solution Components (mL)	2	3	4	5	6	8	10	12
ddH ₂ O	1.4	2.1	2.7	3.4	4.1	5.5	6.8	8.1
30% Acrylamide Solution	0.33	0.5	0.67	0.83	1	1.3	1.7	2.01
Tris (pH 8.8)	0.25	0.38	0.5	0.63	0.75	1	1.25	1.5
10% SDS	0.02	0.03	0.04	0.05	0.06	0.08	0.1	0.12
10% Ammonium Persulfate (APS)	0.02	0.03	0.04	0.05	0.06	0.08	0.1	0.12
TEMED	0.002	0.003	0.004	0.005	0.006	0.008	0.01	0.012

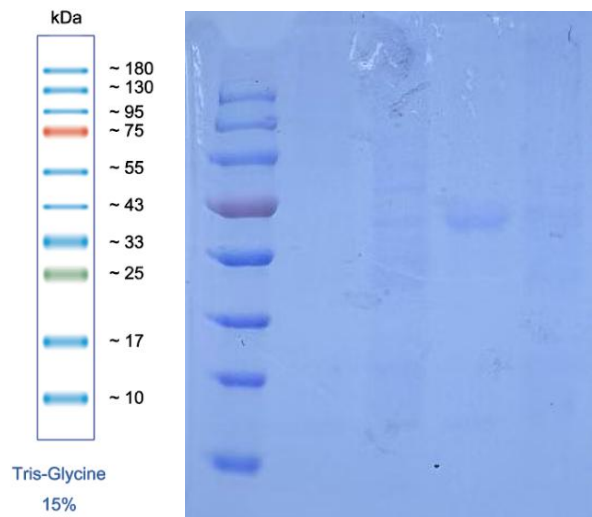
3.electrophoresis

Connect the power supply and run the gel at 80 V for 30 minutes, then run it at 120 V for about 1 hour

4.Dyeing and decoloring

Gently remove the gel from the water, then place it in the staining solution. Boil the solution and incubate the gel with heated staining liquid on a small shaker at room temperature for approximately 45 minutes. Rinse the gel thoroughly with distilled water to remove residual

staining. Add protein decolorizing solution, heat it to boiling, and decolorize the heated solution on a small shaker at room temperature for 30 minutes. The results are shown in the figures:



The results showed that Vdh enzyme protein successfully

Module 2:

1. Inoculation of control groups

Original strains: BW Δ *codA*-pYB1a-*odm*^{WT}

Cultivation in a shaker at 37°C for 12 hours

2. Induction of BW Δ *codA*-pYB1a-*odm*WT, and selected isolated colonies.

Induction System Component	Volume
ZY	4.8 mL
50x 5052 supplement	100 μ L
50x M supplement	100 μ L
MgSO ₄	10 μ L
1000x trace element	10 μ L
Bacterial suspension	50 μ L
20% arabinose	50 μ L
Antibiotic	5 μ L

Cultivation in a shaker at 30°C for 16 hours.

3. Whole-Cell Catalysis

Using 20%Ara to catalyze the induced enzymes for 4-HB production. Determine the OD₆₀₀ of induced bacterial cells using an ultraviolet spectrophotometer. Collect OD₆₀₀=8 units of bacterial cells into a centrifuge tube, with 3 biological replicates for each strain, and centrifuged at 4200 rpm for 10 minutes. Discard the supernatant, resuspend the bacterial cells in 2.5 mM VA at a final concentration, with Tris-HCl (pH = 7.5) as the solvent, and transfer to a centrifuge tube for whole-cell catalysis. Incubate in a shaker at 30°C for 12 hours. After catalysis, the system was centrifuged, filtered through 0.22 μ m membranes, and stored at -20°C.

4. High-Performance Liquid Chromatography (HPLC) Detection of VA

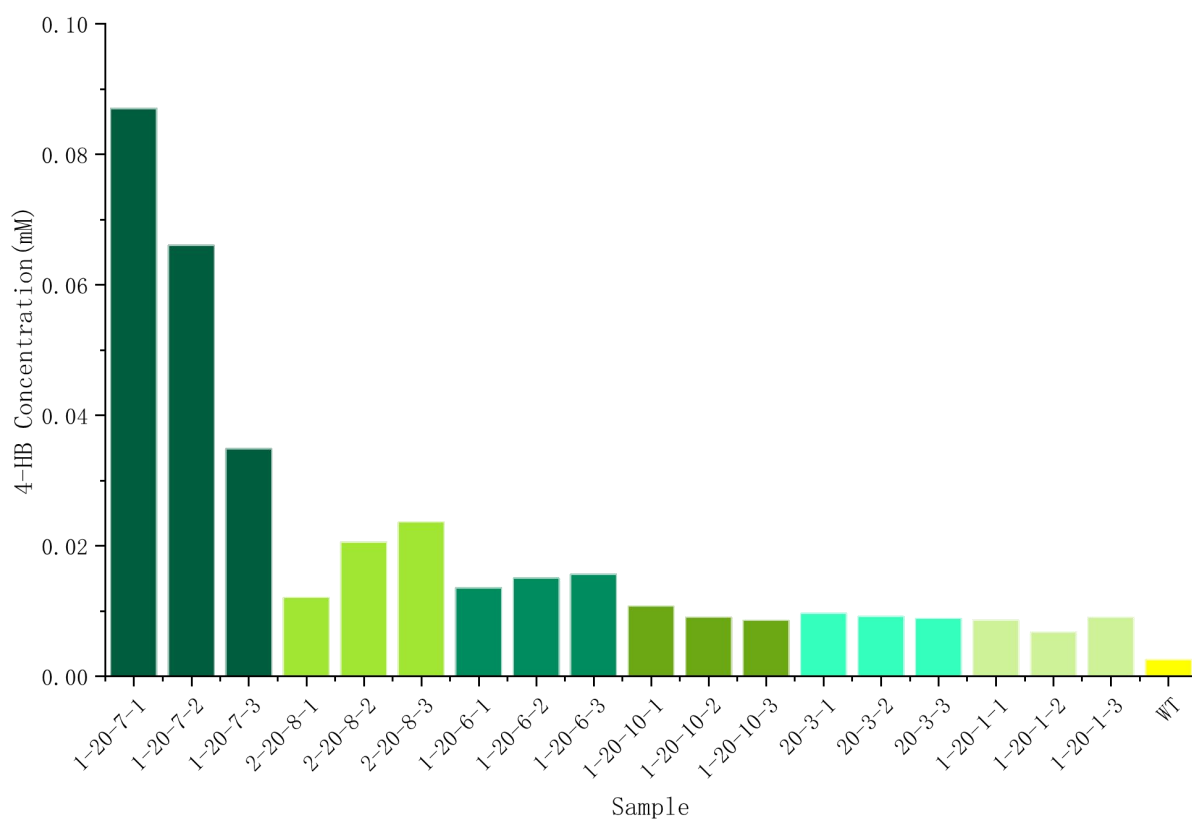
Prepare mobile phase, standards and filter samples

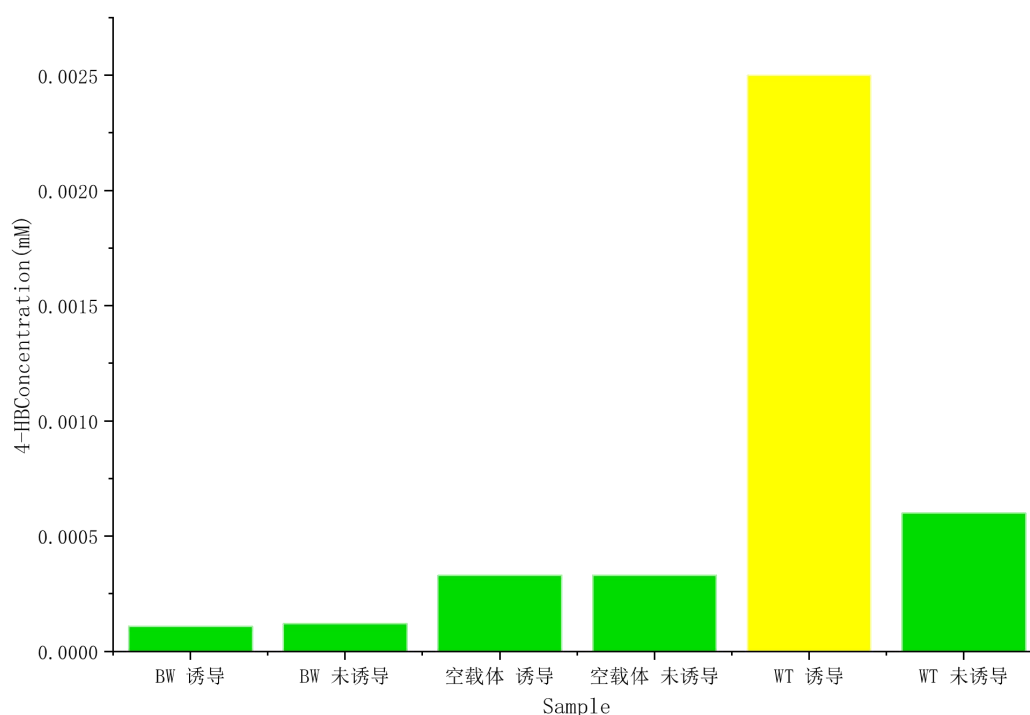
The mobile phase was prepared: 0.1% aqueous phosphoric acid : methanol = 75:25 (v/v)

The standard solutions were prepared, and the samples were filtered through 0.22 μ m membranes for HPLC.

The samples were detected by HPLC using the mobile phase of 0.1% aqueous phosphoric acid : methanol = 75:25 (v/v) . Column: C18,250nm $\times$ 4.6mm $\times$ 5 μ m; flow rate: 0.7 mL/min; column injection volume: 5 μ L;detection wavelength:254nm.

The samples were detected by HPLC using column: C18,250nm $\times$ 4.6mm $\times$ 5 μ m; flow rate: 0.7 mL/min; column injection volume: 5 μ L;detection wavelength:254nm.





High-performance strains exhibited the following 4-HB yields:

1-20-7: 0.0732 mM

2-20-8: 0.0561 mM

1-20-6: 0.0147 mM

1-20-10: 0.0094 mM

20-3: 0.0091 mM

1-20-1: 0.0080 mM

These six high-performance strains showed **29.28-fold, 22.44-fold, 5.88-fold, 3.76-fold, 3.64-fold and 3.20-fold** 4-HB yields compared with the wild-type pYB1a-*odm*^{WT} (0.0025 mM). All concentrations represent the average of three biological replicates and were calculated based on the standard curve.

5. Sequencing

The sequencing results for the best-performance mutant 1-20-7 showed that five mutation sites were identified, including one synonymous mutation: Glu16Gly, Glu49Gly, Arg107Cys and Ala246Thr.

Module 3:

(A)Run HPLC to verify pyruvate production.

1.Shake culture of strains No.1, 2 and 12, and send for testing

Strains No.1, No.2 and No.12 were inoculated and cultivated in LB medium (37°C, 12 hours), and then submitted for sequencing/testing.

2.Culture preservation and plasmid extraction

The strains were preserved (glycerol stock), and plasmids were extracted and their concentrations were measured.

3.Induction: Induction of BW-pLB1s-*VanAB-aroY*

Induction System Component	Volume
ZY	4.8 mL
50x 5052 supplement	100 µL
50x M supplement	100 µL
MgSO4	10 µL
1000x trace element	10 µL
Bacterial suspension	50 µL
arabinose	50 µL
Antibiotic	5 µL

cultivation in a shaker at 30°C for 16 hours.

4.Whole-cell catalysis

Measurement of the OD600 value.

BW-pYB1a-dmpB-dmpD-mhpD-mhpE using VA to catalyze the induced enzymes for Pyr production.BW-pLB1s-VanAB-aroY,BW-pYB1a-dmpB-dmpD-mhpD-mhpE using CA to catalyze the induced enzymes for Pyr production . Determine the OD 600 of induced bacterial cells using an ultraviolet spectrophotometer.Collect OD 600=6 units of bacterial cells into a centrifuge tube, centrifuge at 4200 rpm for 10 minutes.Discard the supernatant, resuspend the bacterial cells in H2SO4, and transfer to a centrifuge tube for whole-cell catalysis. Incubate in a shaker at 30 ° C for 1hours.

The whole-cell catalytic system was configured as follows:

System	Substrate (final conc.)	Cofactors (final conc.)	Solvent
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System	Substrate (final conc.)	Cofactors (final conc.)	Solvent
Uninduced Full pathway (BW-pLB1s-VanAB-aro Y/BW-pYB1a-dmpBD-m hpDE)	VA 5mM (final)	/	5mM H2SO4 solution
Uninduced Full pathway (BW-pLB1s-VanAB-aro Y/BW-pYB1a-dmpBD-m hpDE)	VA 5mM (final)	NADH 10mM (final)	5mM H2SO4 solution
Induced Full pathway (BW-pLB1s-VanAB-aro Y/BW-pYB1a-dmpBD-m hpDE))	VA 5mM (final)	/	5mM H2SO4 solution
Induced Full pathway (BW-pLB1s-VanAB-aro Y/BW-pYB1a-dmpBD-m hpDE))	VA 5mM (final)	NADH 10mM (final)	5mM H2SO4 solution
Uninduced BW	VA 5mM (final)	NADH 10mM (final)	5mM H2SO4 solution
Induced BW	VA 5mM (final)	NADH 10mM (final)	5mM H2SO4 solution
Uninduced BW-pYB1a- <i>eGFP</i>	VA 5mM (final)	NADH 10mM (final)	5mM H2SO4 solution
Induced BW-pYB1a- <i>eGFP</i>	VA 5mM (final)	NADH 10mM (final)	5mM H2SO4 solution
Uninduced Downstream pathway (BW-pYB1a-dmpBD-mh pDE)	CA 5mM (final)	/	5mM H2SO4 solution
Uninduced Downstream pathway (BW-pYB1a-dmpBD-mh pDE)	CA 5mM (final)	Fe2+ 0.5mM (final), Mn2+ 1mM (final)	5mM H2SO4 solution
Induced Downstream pathway (BW-pYB1a-dmpBD-mh pDE)	CA 5mM (final)	/	5mM H2SO4 solution
Induced Downstream pathway (BW-pYB1a-dmpBD-mh pDE)	CA 5mM (final)	Fe2+ 0.5mM (final), Mn2+ 1mM (final)	5mM H2SO4 solution
Uninduced BW	CA 5mM (final)	Fe2+ 0.5mM (final), Mn2+ 1mM (final)	5mM H2SO4 solution

System	Substrate (final conc.)	Cofactors (final conc.)	Solvent
Induced BW	CA 5mM (final)	Fe2+ 0.5mM (final), Mn2+ 1mM (final)	5mM H2SO4 solution
Uninduced BW-pYB1a- <i>eGFP</i>	CA 5mM (final)	Fe2+ 0.5mM (final), Mn2+ 1mM (final)	5mM H2SO4 solution
Induced BW-pYB1a- <i>eGFP</i>	CA 5mM (final)	Fe2+ 0.5mM (final), Mn2+ 1mM (final)	5mM H2SO4 solution

All components were added to the cell pellet tubes and mixed by pipetting, then shaken at 30°C, 200r/min in the dark.

5.Prepare standards and mobile phase

The HPLC standard solutions and the mobile phase (5mM H2SO4 aqueous solution, etc.) were prepared.

6.HPLC (acetonitrile-ultrapure water method; 5mM dilute H2SO4 sugar-H column method)

The samples were detected by HPLC with the acetonitrile-ultrapure water method (mobile phases: 5% acetonitrile/5mM H2SO4, 30% acetonitrile/5mM H2SO4)

7.HPLC run.

The standards and the samples (full pathway and downstream pathway) were detected by HPLC with RID.

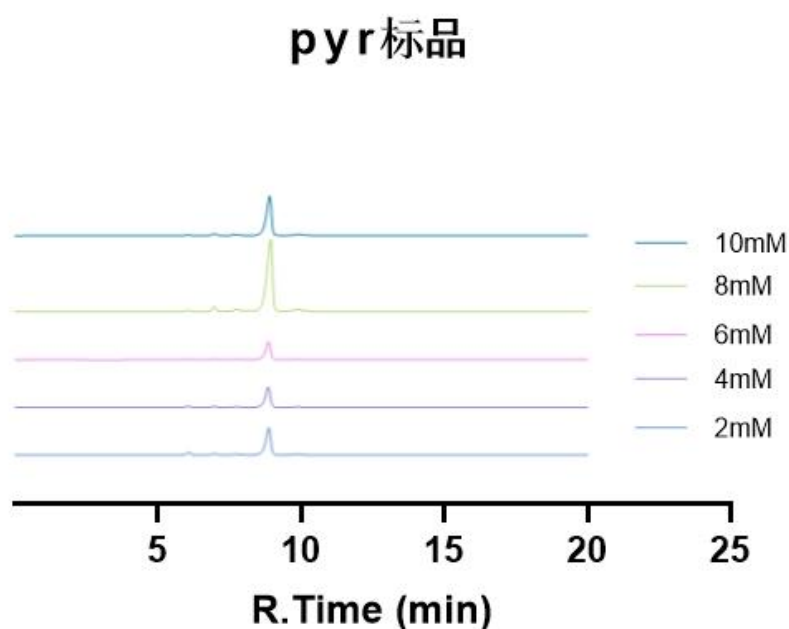
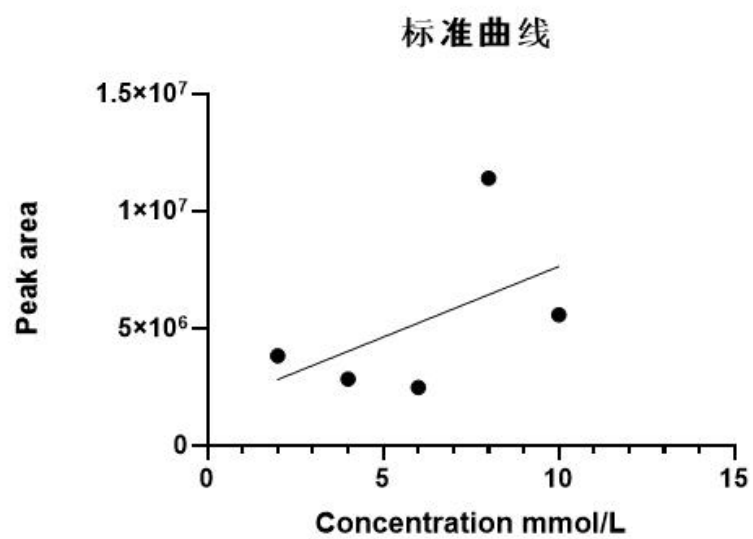
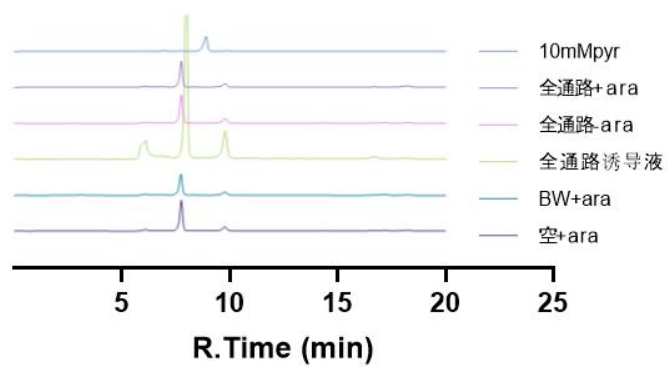


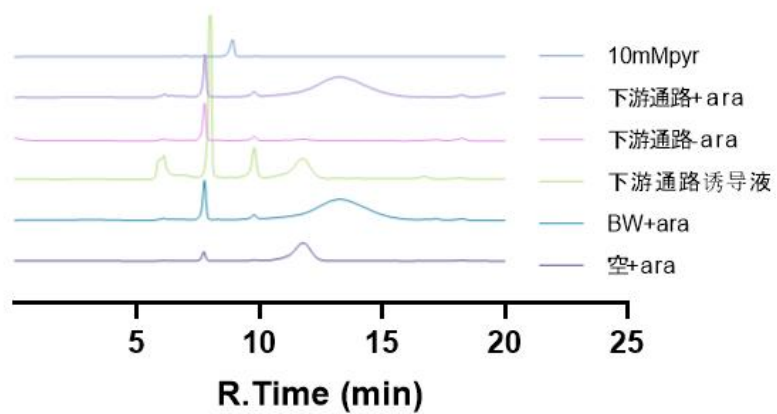
Figure 9: HPLC chromatograms of the pyruvate standards (10/8/6/4/2 mM).



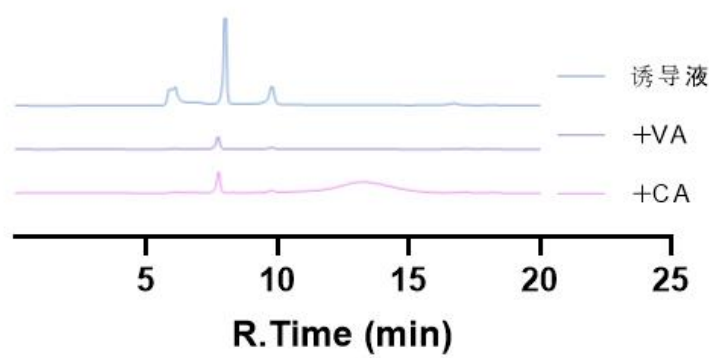
pLB1s-VanAB-aroY/pYB1a-dmpBD-mhpDE/BW



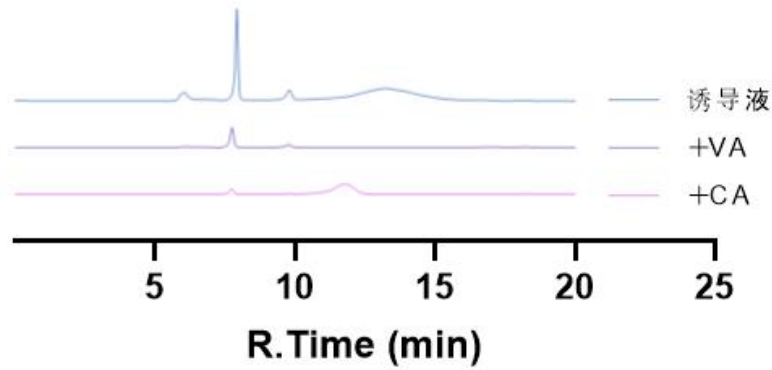
pYB1a-dmpBD-mhpDE/BW



BW



pYB1a-eGFP



(B)RBS library construction (dmpB deletion and RBS mutation).

1.dmpD single-gene PCR construction.

The low-fidelity PCR was performed to obtain the single fragment, and the product was checked by running a recovery gel.

2.Gel recovery, ligation, transformation and plating.

Gel recovery and ligation were performed, followed by transformation and plating; gel recovery, overlap extension, Gibson ligation, transformation and plating were also carried out.

3.Plasmid extraction for sequencing.

The RBS-library plasmids were extracted and submitted for sequencing (15 plasmids were extracted).

4.RBS library construction - dmpB deletion and RBS mutation.

Since the primer for detecting the mhpD mutation site was designed on the dmpB gene upstream of it, the deletion of dmpB caused the sequencing report to fail.

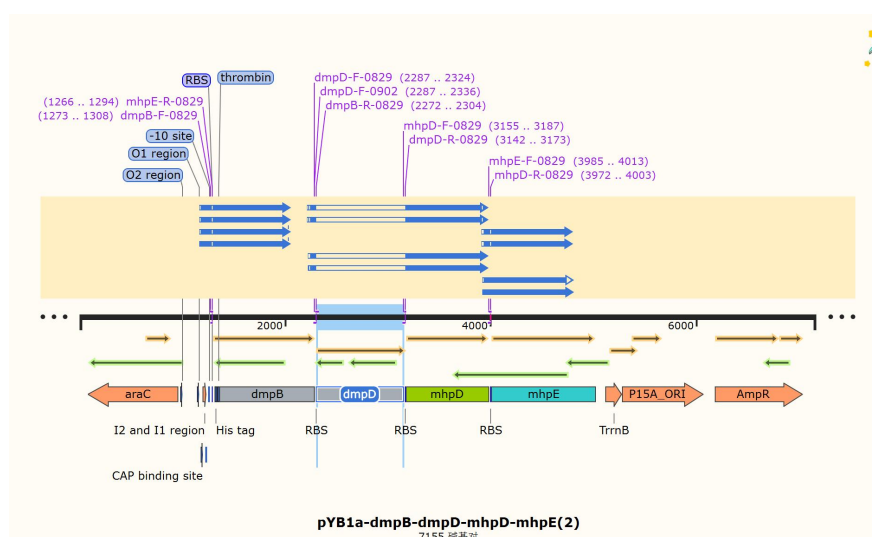


Figure 15: Plasmid map of pYB1a-dmpB-dmpD-mhpD-mhpE(2) (7155 bp).

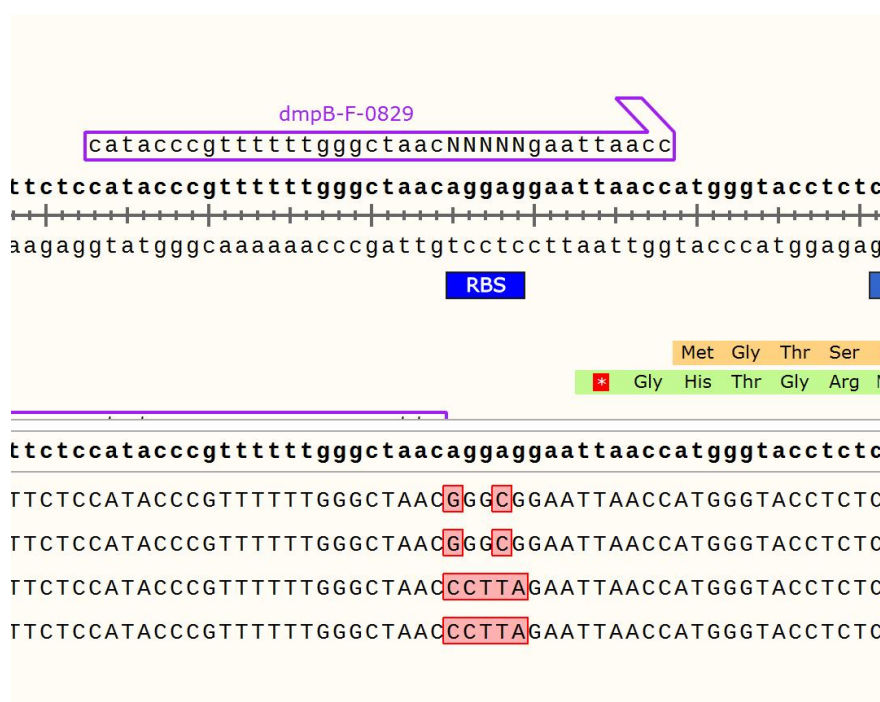


Figure 16: Sequence comparison around the dmpB RBS with the primer dmpB-F-0829.

Group	Component	Amount
A (60mL plain 1×M9, antibiotic-free)	5×M9 salt solution	12.00 mL
	20% glycerol stock	150.00 µL
	1M MgSO ₄	120.00 µL
	Sterile ddH ₂ O	47.77 mL
	1M CaCl ₂	6.00 µL
B (40mL 1×M9 with 5mM VA, 10mM Fe ²⁺ , 10mM Mn ²⁺ , AS)	Neutral vanillic acid solution (10mM VA stock 20.00 mL + sterile ddH ₂ O 11.01 mL + 1M NaOH to neutral pH)	—
	5×M9 salt solution	8.00 mL
	20% glycerol stock	99.69 µL
	1M MgSO ₄	80.00 µL
	1M FeSO ₄	400.00 µL
	1M MnSO ₄	400.00 µL
	AS antibiotic stock	40.00 µL
	1M CaCl ₂	4.00 µL
	10 g/L sodium pyruvate stock	18.00 mL
	Sterile ddH ₂ O	29.66 mL
C (60mL 1×M9 with sodium pyruvate, antibiotic-free)	5×M9 salt solution	12.00 mL
	20% glycerol stock	149.54 µL
	1M MgSO ₄	120.00 µL
	1M CaCl ₂	6.00 µL

Addition system: 8 groups in triplicate were set up. Strain ①: RBS mutant; Strain ②: wild type without RBS mutation; Strain ③: pyr-knockout sensor strain.

Addition system (each group in triplicate):

No.	Medium	Strain
1	A blank 1×M9	Strain ① (RBS mutant) + AS
		Strain ② (wild type, no RBS mutation) + AS
2	A blank 1×M9	

No.	Medium	Strain
3	A blank 1×M9	Strain ③ (pyr-knockout sensor)
4	B (with VA, Fe ²⁺ , Mn ²⁺)	Strain ① (RBS mutant)
5	B (with VA, Fe ²⁺ , Mn ²⁺)	Strain ② (wild type)
6	C (with 3mM sodium pyruvate)	Strain ① (RBS mutant) + AS
7	C (with 3mM sodium pyruvate)	Strain ② (wild type) + AS
8	C (with 3mM sodium pyruvate)	Strain ③ (pyr-knockout sensor)

Week 17 (9.12-9.17)

Module 1:

1. Inoculation

Original strains: BW-pYB1a-*progo*, BW-pYB1a-*nov1f*, BW-pYB1a-*progo-nov1f*, BW-pYB1a-*vdh*, BW-pYB1a-*vdh-nox*, BW-pRB1k-*vdh-nox*

Cultivation in a shaker at 37°C for 12 hours

2. Induction:

Induction of BW-pYB1a-*progo*, BW-pYB1a-*nov1f*, BW-pYB1a-*progo-nov1f*, BW-pYB1a-*vdh*, BW-pYB1a-*vdh-nox*, BW-pRB1k-*vdh-nox*

Induction System Component	Volume
ZY	4.8 mL
50x 5052 supplement	100 µL
50x M supplement	100 µL
MgSO ₄	10 µL

1000x trace element	10 μ L
Bacterial suspension	50 μ L
arabinose	50 μ L
Antibiotic	5 μ L

cultivation in a shaker at 30°C for 16 hours.

3.1.1 Whole-Cell Catalysis:

Using vanillin to catalyze the induced enzymes for VA production

Determine the OD₆₀₀ of induced bacterial cells using an ultraviolet spectrophotometer.

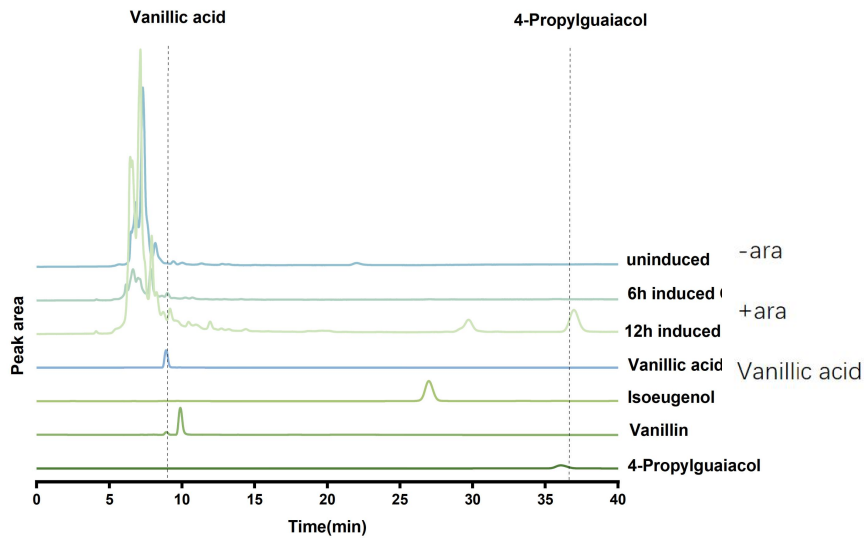
Collect 6 OD units of bacterial cells into a centrifuge tube, centrifuge at 4200 rpm for 10 minutes.

Discard the supernatant, resuspend the bacterial cells in 50mM Tris-HCl(pH=7.5), and transfer to a centrifuge tube for whole-cell catalysis.

Incubate in a shaker at 30°C for 6h ,12h hours.

3.1.2 High-Performance Liquid Chromatography (HPLC)

Detection of VA



The results showed that under the conditions of 5MM 4-PG, the yields at 6 hours 9 OD units of bacterial cells and at 12 hours 6 OD units of bacterial cells are similar, approximately 0.47%.

3.2 Protein expression validation

1. Sample Preparation

The cultures were collected, and the bacterial suspension with a 6 OD unit was measured to obtain the final sample. After centrifugation at 10,000 g for 10 minutes, the supernatant was discarded, and the precipitate was resuspended in 600 μ L sterile water. The cells were lysed using an ultrasonic cell disruptor. The lysate was then centrifuged at 10,000 g for 10 minutes to separate the supernatant and sediment. Finally, the precipitate was resuspended in 600 μ L sterile water. This procedure completed the sample preparation.

2. point sample

Prepare the protein buffer solution according to the table below. Place the gel in the electrophoresis instrument. After checking for leaks with the buffer solution, mix 16 μL of prepared sample with 4 μL loading buffer. Transfer the mixed sample into the PCR instrument and heat at 100°C for 5 minutes to denature proteins. After cooling, load 16 μL of the mixture into the gel well and add 4 μL of protein marker. The sample loading is now complete.

Concentrating Gel:

12% Lower Gel	2 Gel Units	3 Gel Units	4 Gel Units	5 Gel Units	6 Gel Units	8 Gel Units	10 Gel Units
Solution Components (mL)	10	15	20	25	30	40	50
Water	3.3	4.9	6.6	8.2	9.9	13.2	16.5
30% Acrylamide Solution	4	6	8	10	12	16	20
Tris (pH 8.8)	2.5	3.8	5	6.3	7.5	10	12.5
10% SDS	0.1	0.15	0.2	0.25	0.3	0.4	0.5
10% Ammonium Persulfate	0.1	0.15	0.2	0.25	0.3	0.4	0.5
TEMED	0.004	0.006	0.008	0.01	0.012	0.016	0.02

separation gel:

5% Stacking Gel	2 Gel Units	3 Gel Units	4 Gel Units	5 Gel Units	6 Gel Units	8 Gel Units	10 Gel Units	12 Gel Units
Solution Components (mL)	2	3	4	5	6	8	10	12
ddH ₂ O	1.4	2.1	2.7	3.4	4.1	5.5	6.8	8.1
30% Acrylamide Solution	0.33	0.5	0.67	0.83	1	1.3	1.7	2.01
Tris (pH 8.8)	0.25	0.38	0.5	0.63	0.75	1	1.25	1.5
10% SDS	0.02	0.03	0.04	0.05	0.06	0.08	0.1	0.12
10% Ammonium Persulfate (APS)	0.02	0.03	0.04	0.05	0.06	0.08	0.1	0.12
TEMED	0.002	0.003	0.004	0.005	0.006	0.008	0.01	0.012

3.electrophoresis

Connect the power supply and run the gel at 80 V for 30 minutes, then run it at 120 V for about 1 hour

4. Dyeing and decoloring

Gently remove the gel from the water, then place it in the staining solution. Boil the solution and incubate the gel with heated staining liquid on a small shaker at room temperature for approximately 45 minutes. Rinse the gel thoroughly with distilled water to remove residual staining. Add protein decolorizing solution, heat it to boiling, and decolorize the heated solution on a small shaker at room temperature for 30 minutes.



The preparation time of ammonium persulfate is too long.